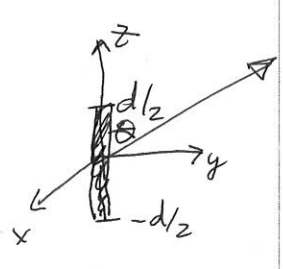
Linear, sinusoidal, centre-fed antenna



$$\vec{J}(\vec{r}) = \begin{cases} I \sin\left(\frac{kd}{2} - k|z|\right) \delta(x)\delta(y)\, \hat{e}_z, \\ \text{for } |z| < \frac{d}{2}; \\ 0 \text{ for } |z| > \frac{d}{2} \end{cases} \quad \left(\begin{matrix}\text{standing current} \\ \text{wave}\end{matrix}\right)$$

From <u>Lect. 3A</u> $\vec{B}(r) = ik\,\hat{n}\times\vec{A} = ik\,\frac{\mu_0}{4\pi r}\, e^{ikr} \int \hat{n}\times\vec{J}(\vec{r}\,')\, e^{-ik\hat{n}\cdot\vec{r}\,'}\, d\vec{r}\,'$

$\hat{n} = \frac{\vec{r}}{r} = \hat{r}$ is the unit vector to observation point.

let $\hat{n}\cdot\hat{e}_z = \cos\theta$

$\cancel{\vec{B} \propto \frac{ik\mu_0 e^{ikr}}{4\pi r}}$

$$\boxed{\text{Remember that these are expressions for } e^{-i\omega t} \text{ components of } \vec{A}(\vec{r},t),\ \vec{B}(\vec{r},t),\ \vec{E}(\vec{r},t)}$$

$$\vec{A} = \frac{\mu_0}{4\pi r}\, e^{ikr} \int_{-d/2}^{d/2} dz'\, I \sin\left(\frac{kd}{2} - k|z'|\right) e^{-ikz'\cos\theta}\, \hat{e}_z$$

$$= \frac{\mu_0 I e^{ikr}}{4\pi r}\, \hat{e}_z\, \underbrace{\frac{1}{2i}\int_{-d/2}^{d/2} dz' \left(e^{i\frac{kd}{2} - ik|z'|} - e^{-i\frac{kd}{2} + ik|z'|}\right) e^{-ikz'\cos\theta}}$$

denote $C = \frac{1}{2i}\left\{ \int_0^{d/2} dz'\, e^{ikd/2}\, e^{-ikz'}\, e^{-ikz'\cos\theta} + e^{i\frac{kd}{2}} \int_{-d/2}^{0} dz'\, e^{ikz'}\, e^{-ikz'\cos\theta} \right.$

$$\left. - e^{-i\frac{kd}{2}} \left(\int_0^{d/2} dz'\, e^{ikz' - ikz'\cos\theta} + \int_{-d/2}^{0} dz'\, e^{-ikz' - ikz'\cos\theta} \right) \right\}$$

$$= \frac{1}{2i}\left\{ e^{i\frac{kd}{2}} \left(\left.\frac{e^{-ik(1+\cos\theta)z'}}{-ik(1+\cos\theta)}\right|_0^{d/2} + \left.\frac{e^{ik(1-\cos\theta)z'}}{ik(1-\cos\theta)}\right|_{-d/2}^{0} \right) \right.$$

$$\left. - e^{-i\frac{kd}{2}} \left(\left.\frac{e^{ik(1-\cos\theta)z'}}{ik(1-\cos\theta)}\right|_0^{d/2} + \left.\frac{e^{-ik(1+\cos\theta)z'}}{-ik(1+\cos\theta)}\right|_{-d/2}^{0} \right) \right\}$$

$$C = \frac{1}{2i}\left\{ e^{ikd/2}\left(\frac{e^{-ik\frac{d}{2}(1+\cos\theta)} - 1}{-ik(1+\cos\theta)} + \frac{1 - e^{-ik\frac{d}{2}(1-\cos\theta)}}{ik(1-\cos\theta)} \right) \right.$$

$$\left. - e^{-ikd/2}\left(\frac{e^{ik\frac{d}{2}(1-\cos\theta)} - 1}{ik(1-\cos\theta)} + \frac{1 - e^{ik\frac{d}{2}(1+\cos\theta)}}{-ik(1+\cos\theta)} \right) \right\} =$$

$$= \frac{1}{2i}\left\{ \frac{e^{-ikd\cos\theta/2} - e^{ikd/2}}{-ik(1+\cos\theta)} + \frac{e^{ikd/2} - e^{ikd\cos\theta/2}}{ik(1-\cos\theta)} \right.$$

$$\left. - \left(\frac{e^{-ikd\cos\theta/2} - e^{-ikd/2}}{ik(1-\cos\theta)} + \frac{e^{-ikd/2} - e^{ikd\cos\theta/2}}{-ik(\cancel{k+}\cos\theta)} \right) \right\}$$

$$= \frac{1}{2i}\left\{ \frac{e^{-ikd\cos\theta/2} + e^{ikd\cos\theta/2} - e^{ikd/2} - e^{-ikd/2}}{-ik(1+\cos\theta)} \right.$$

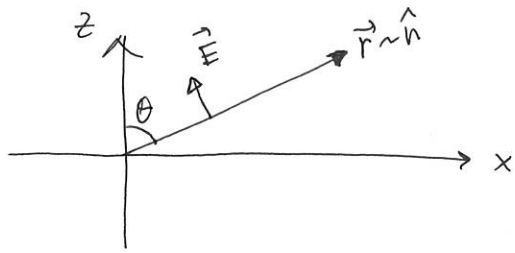
$$\left. + \frac{e^{ikd/2} + e^{-ikd/2} - e^{ikd\cos\theta/2} - e^{-ikd\cos\theta/2}}{ik(1-\cos\theta)} \right\}$$

$$= \frac{2\left[\cos(kd\cos\theta/2) - \cos(kd/2)\right]}{2k(1+\cos\theta)} + \frac{2\left[\cos\left(\frac{kd}{2}\right) - \cos(kd\cos\theta/2)\right]}{-2k(1-\cos\theta)}$$

$$= \frac{\cos\left(\frac{kd}{2}\cos\theta\right) - \cos\left(\frac{kd}{2}\right)}{k} \underbrace{\left(\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} \right)}_{\frac{2}{1-\cos^2\theta} = \frac{2}{\sin^2\theta}}$$

$$\vec{A} = \hat{e}_z \left\{ \frac{\mu_0 I e^{ikr}}{4\pi r} \frac{2\left[\cos\left(\frac{kd}{2}\cos\theta\right) - \cos\left(\frac{kd}{2}\right)\right]}{k\sin^2\theta} \right\}$$

$$\Rightarrow \vec{B} = ik\hat{n}\times\hat{e}_z\{\quad\}, \quad \vec{E} = c\vec{B}\times\hat{n} = ick\underbrace{\left(\hat{n}\times\hat{e}_z\times\hat{n}\right)}_{\hat{e}_z - \cos\theta\,\hat{n}}\{\quad\}$$



$\hat{e}_z - \cos\theta\, \hat{n} \perp \hat{n}$

since $(\hat{e}_z - \cos\theta\, \hat{n}) \cdot \hat{n} = 0$

and $\vec{B} \perp \vec{E}$, $\vec{B} \perp \hat{n}$

Pointing vector

$$\vec{S}(\vec{r},t) = \frac{1}{\mu_0} \vec{E}(\vec{r},t) \times \vec{B}(\vec{r},t)$$

Physical fields $\vec{E}_{phys}(\vec{r},t) = \mathrm{Re}\left\{ \vec{E}(\vec{r})\, e^{-i\omega t} \right\} =$

$$= \frac{1}{2}\left(\vec{E}(\vec{r})\, e^{-i\omega t} + \vec{E}^*(\vec{r})\, e^{i\omega t} \right)$$

$$\vec{B}_{phys}(\vec{r},t) = \mathrm{Re}\left\{ \vec{B}(\vec{r})\, e^{-i\omega t} \right\}.$$

Then $\vec{S}(\vec{r},t) = \frac{1}{\mu_0} \frac{1}{2}\left(\vec{E}(\vec{r})\, e^{-i\omega t} + \vec{E}^*(\vec{r})\, e^{i\omega t} \right) \times$

$$\times \frac{1}{2}\left(\vec{B}(\vec{r})\, e^{-i\omega t} + \vec{B}^*(\vec{r})\, e^{i\omega t} \right)$$

$$= \frac{1}{4\mu_0}\Big(\vec{E}(\vec{r}) \times \vec{B}^*(\vec{r}) + \vec{E}^*(\vec{r}) \times \vec{B}(\vec{r}) + \vec{E}(\vec{r}) \times \vec{B}(\vec{r})\, e^{-i2\omega t}$$

$$+ \vec{E}^*(\vec{r}) \times \vec{B}(\vec{r})\, e^{i2\omega t} \Big)$$

This is instantaneous value of $\vec{S}(\vec{r},t)$.

Let us consider more meaningful — time averaged:

$$\langle \vec{S}(\vec{r},t) \rangle = \frac{1}{T} \int_0^T \vec{S}(\vec{r},t)\, dt = \frac{1}{2\pi} \int_0^{\omega T = 2\pi} d(\omega t)\, \vec{S}(\vec{r},t) = \frac{1}{4\mu_0} \cdot \frac{1}{2\pi} \int_0^{2\pi} dz \cdot$$

$T = \frac{2\pi}{\omega}$

$$\cdot \left(\vec{E} \times \vec{B}^* + \vec{E}^* \times \vec{B} + \vec{E} \times \vec{B}\, e^{-i2z} + \vec{E}^* \times \vec{B}^*\, e^{i2z} \right)$$

But $\int_0^{2\pi} dz\, e^{i2z} = \left. \frac{e^{i2z}}{2i} \right|_0^{2\pi} = \frac{e^{i4\pi} - 1}{2i} = 0$

Thus $\langle \vec{S}(\vec{r},t)\rangle = \frac{1}{4\mu_0}\left(\vec{E}(r)\times\vec{B}^*(r) + \vec{E}^*(r)\times\vec{B}(r)\right)$

$$\vec{E}\times\vec{B}^* \sim \left(\hat{e}_z - \cos\theta\,\hat{n}\right)\times\left(\hat{n}\times\hat{e}_z\right) = \hat{e}_z\times\hat{n}\times\hat{e}_z$$

$$-\cos\theta\;\hat{n}\times(\hat{n}\times\hat{e}_z) = \hat{n}(\hat{e}_z\cdot\hat{e}_z) - \hat{e}_z(\hat{n}\cdot\hat{e}_z)$$

$$-\cos\theta\left(\hat{n}(\hat{n}\cdot\hat{e}_z) - \hat{e}_z(\hat{n}\cdot\hat{n})\right) =$$

$$= \hat{n} - \cos\theta\,\hat{e}_z - \cos^2\theta\,\hat{n} + \cos\theta\,\hat{e}_z$$

$$= \sin^2\theta\,\hat{n}.$$

Hence $\langle \vec{S}(\vec{r},t)\rangle = \frac{ck^2}{2\mu_0}\sin^2\theta\,\hat{n}\left(\frac{\mu_0 I}{2\pi}\right)^2\left(\frac{\cos\left(\frac{kd}{2}\cos\theta\right)-\cos\frac{kd}{2}}{kr\sin^2\theta}\right)^2$

$$= \frac{cI^2}{8\pi^2\mu_0}\hat{n}\left(\frac{\cos\left[\frac{kd}{2}\cos\theta\right]-\cos\left[\frac{kd}{2}\right]}{r\,\sin\theta}\right)^2.$$

Power radiated into spherical angle $d\Omega$: area $d\vec{f} = r^2 d\Omega\cdot\hat{n}$

$$dP = \langle\vec{S}\rangle\cdot d\vec{f} = |\langle\vec{S}\rangle|\,r^2\,d\Omega$$

$$\Rightarrow \frac{dP}{d\Omega} = \frac{cI^2}{8\pi^2\mu_0}\left(\frac{\cos\left[\frac{kd}{2}\cos\theta\right]-\cos\left[\frac{kd}{2}\right]}{\sin\theta}\right)^2$$

is independent of r! Just because $\frac{1}{r^2}\cdot r^2 = 1$.

Investigate limits of $\frac{dP}{d\Omega}$ as function of kd:

1) $kd \ll 1 \Rightarrow \frac{d}{\lambda} \ll 1$: size of antenna is much smaller than wavelength (long wavelength limit)

$$\cos\left(\frac{kd}{2}\cos\theta\right) - \cos\left(\frac{kd}{2}\right) = 1 - \left(\frac{kd}{2}\cos\theta\right)^2 - \left(1 - \left(\frac{kd}{2}\right)^2\right) =$$

$$= \left(\frac{kd}{2}\right)^2 (1 - \cos^2\theta) = \left(\frac{kd}{2}\right)^2 \sin^2\theta$$

$$\frac{dP}{d\Omega} = \frac{cI^2}{32\pi^2\mu_0}\left(\frac{kd}{2}\right)^4 \sin^2\theta \quad \begin{cases} \text{peaked at } \theta = \pi/2 \,! \\ \sim \omega^4 \end{cases}$$

Can be written in terms of $I_0 = I\frac{kd}{2}$ = peak current of the antenna

$$\vec{J}(\vec{r}) = I \sin\left(\frac{kd}{2} - k|z|\right)\hat{e}_z \approx \underbrace{I\frac{kd}{2}}_{I_0}\left(1 - \frac{2|z|}{d}\right)\hat{e}_z$$

for $|z| < \frac{d}{2}$.

$$\frac{dP}{d\Omega} = \frac{c\, I_0^2}{32\pi^2\mu_0}(kd)^2 \sin^2\theta$$

$$k = \frac{\omega}{c}$$

2) $kd \gg 1$: then $\cos\left(\frac{kd}{2}\cos\theta\right) - \cos\left(\frac{kd}{2}\right)$ oscillates frequently with $\cos\theta$: very different radiation profile.